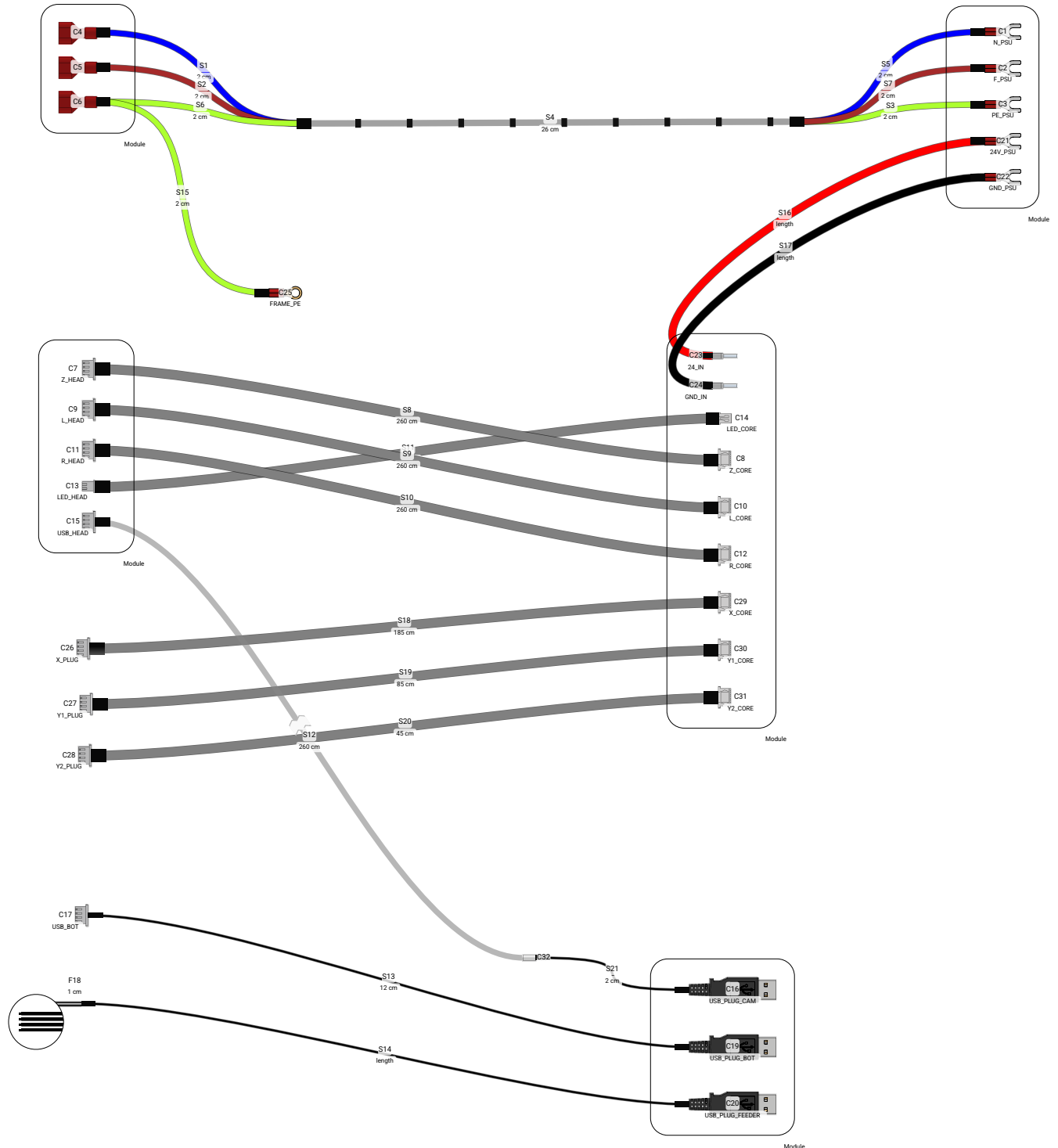


## Description

Harness from PCB to devices and toolboard



## Connectors

| Letter | Name                                                                   | Manufacturer<br>PN | Custom Name     | Number of<br>Pins | Gender |
|--------|------------------------------------------------------------------------|--------------------|-----------------|-------------------|--------|
| C1     | M5 Crimp Fork Terminal Red 16-22AWG                                    | 534-250            | N_PSU           |                   |        |
| C3     | M5 Crimp Fork Terminal Red 16-22AWG                                    | 534-250            | PE_PSU          |                   |        |
| C2     | M5 Crimp Fork Terminal Red 16-22AWG                                    | 534-250            | F_PSU           |                   |        |
| C4     | Crimp 6.3x0.8mm Faston M 22-16AWG Insulated Red                        | BM 80180           |                 |                   |        |
| C5     | Crimp 6.3x0.8mm Faston M 22-16AWG Insulated Red                        | BM 80180           |                 |                   |        |
| C6     | Crimp 6.3x0.8mm Faston M 22-16AWG Insulated Red                        | BM 80180           |                 |                   |        |
| C7     | 4 Pin XH F Shell (JST)                                                 | XHP-4              | Z_HEAD          | 4                 | Female |
| C8     | 4 Pin XH F Shell (JST)                                                 | XHP-4              | Z_CORE          | 4                 | Female |
| C9     | 4 Pin XH F Shell (JST)                                                 | XHP-4              | L_HEAD          | 4                 | Female |
| C10    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | L_CORE          | 4                 | Female |
| C11    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | R_HEAD          | 4                 | Female |
| C12    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | R_CORE          | 4                 | Female |
| C14    | 2 Pin XH F Shell (JST)                                                 | XHP-2              | LED_CORE        | 2                 | Female |
| C13    | 2 Pin XH F Shell (JST)                                                 | XHP-2              | LED_HEAD        | 2                 | Female |
| C15    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | USB_HEAD        | 4                 | Female |
| C16    | 4 Pin USB-A 2.0 M (CNC Tech)                                           | 1002-001-BL-KIT    | USB_PLUG_CAM    | 4                 | Male   |
| C17    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | USB_BOT         | 4                 | Female |
| C19    | 4 Pin USB-A 2.0 M (CNC Tech)                                           | 1002-001-BL-KIT    | USB_PLUG_BOT    | 4                 | Male   |
| C20    | 4 Pin USB-A 2.0 M (CNC Tech)                                           | 1002-001-BL-KIT    | USB_PLUG_FEEDER | 4                 | Male   |
| C21    | M5 Crimp Fork Terminal Red 16-22AWG                                    | 534-250            | 24V_PSU         |                   |        |
| C22    | M5 Crimp Fork Terminal Red 16-22AWG                                    | 534-250            | GND_PSU         |                   |        |
| C23    | Crimp Butt Splice Terminal Blue 10-12AWG to 14-16AWG (TE Connectivity) | 327637             | 24_IN           |                   |        |
| C24    | Crimp Butt Splice Terminal Blue 10-12AWG to 14-16AWG (TE Connectivity) | 327637             | GND_IN          |                   |        |
| C25    | M5 Crimp Ring Terminal Red 15-23AWG                                    | 130014             | FRAME_PE        |                   |        |
| C26    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | X_PLUG          | 4                 | Female |
| C27    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | Y1_PLUG         | 4                 | Female |
| C28    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | Y2_PLUG         | 4                 | Female |
| C29    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | X_CORE          | 4                 | Female |
| C30    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | Y1_CORE         | 4                 | Female |
| C31    | 4 Pin XH F Shell (JST)                                                 | XHP-4              | Y2_CORE         | 4                 | Female |
| C32    | Crimp Butt Splice Uninsulated 22-16AWG                                 | 809-2369           |                 |                   |        |

## Pins

| Letter | Name                          | Manufacturer PN | Gender |
|--------|-------------------------------|-----------------|--------|
| C7     | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C8     | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C9     | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C10    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C11    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C12    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C14    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C13    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C15    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C17    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C26    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C27    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |

| Letter | Name                          | Manufacturer PN | Gender |
|--------|-------------------------------|-----------------|--------|
| C28    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C29    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C30    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |
| C31    | XH F Crimp Pin 22-28AWG (JST) | SXH-001T-P0.6   | Female |

## Wires

| Path | Connectors | Name                                                   | Manufacturer PN | Length | Path Screening | Path Sleeving | Quantity |
|------|------------|--------------------------------------------------------|-----------------|--------|----------------|---------------|----------|
| 1    | C4-C1      | 16AWG 30/0.26mm Blue PVC UL1015 (Eland Cables)         | A2TBL0015       | 30 cm  | None           | None          | 1        |
| 2    | C5-C2      | 16AWG 30/0.26mm Brown PVC UL1015 (Eland Cables)        | A2TBR0015       | 30 cm  | None           | None          | 1        |
| 3    | C6-C3      | 16AWG 30/0.26mm Green/Yellow PVC UL1015 (Eland Cables) | A2TGY0015       | 30 cm  | None           | None          | 1        |
| 5    | C7-C8      | Supertronic PVC                                        |                 | 260 cm | None           | None          | 1        |
| 6    | C9-C10     | Supertronic PVC                                        |                 | 260 cm | None           | None          | 1        |
| 7    | C11-C12    | Supertronic PVC                                        |                 | 260 cm | None           | None          | 1        |
| 8    | C13-C14    | Supertronic PVC                                        |                 | 260 cm | None           | None          | 1        |
| 10   | C17-C19    | 4Core 26AWG White PTFE Foil Screened                   | KAP281 - ZPRKT  | 12 cm  | None           | None          | 1        |
| 11   | F18-C20    | 4Core 26AWG White PTFE Foil Screened                   | KAP281 - ZPRKT  | 1 cm   | None           | None          | 1        |
| 12   | C6-C25     | 16AWG 30/0.26mm Green/Yellow PVC UL1015 (Eland Cables) | A2TGY0015       | 2 cm   | None           | None          | 1        |
| 13   | C23-C21    | 12AWG 56/0.31mm Red Silicone (Donau)                   | 25400           |        | None           | None          | 1        |
| 14   | C24-C22    | 12AWG 56/0.31mm Black Silicone (Donau)                 | 25401           |        | None           | None          | 1        |
| 15   | C26-C29    | Supertronic PVC                                        |                 | 185 cm | None           | None          | 1        |
| 16   | C27-C30    | Supertronic PVC                                        |                 | 85 cm  | None           | None          | 1        |
| 17   | C28-C31    | Supertronic PVC                                        |                 | 45 cm  | None           | None          | 1        |
| 18   | C32-C16    | 4Core 26AWG White PTFE Foil Screened                   | KAP281 - ZPRKT  | 2 cm   | None           | None          | 1        |

## Multicore Screening

| Path | Wire Name                            | Wire Index | Type                         | Screening | Screening     | Screening | Screening     |
|------|--------------------------------------|------------|------------------------------|-----------|---------------|-----------|---------------|
| 10   | 4Core 26AWG White PTFE Foil Screened | 1          | Screened (Foil + Drain Wire) | C17       | to pins (4)   | C19       | to pins (4)   |
| 11   | 4Core 26AWG White PTFE Foil Screened | 1          | Screened (Foil + Drain Wire) | F18       | not connected | C20       | not connected |

## Flying Lead Locations

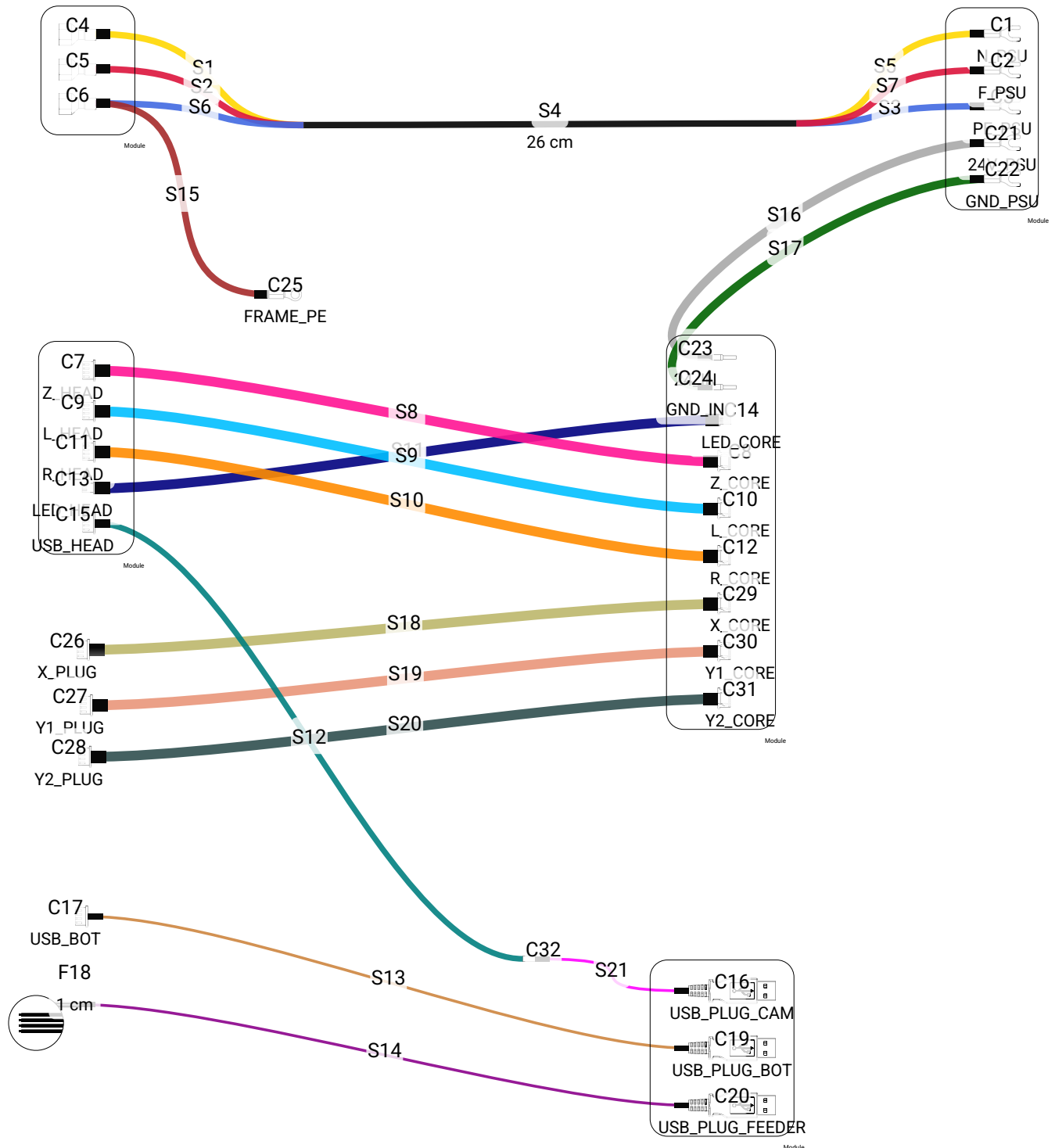
| Letter | Flying lead length | Ends   | Labeling |
|--------|--------------------|--------|----------|
| F18    | 1 cm               | tinned | none     |

## Segments

| Marking | Connectors | Length | Labeling | Bundle Screening | Bundle Sleeve |
|---------|------------|--------|----------|------------------|---------------|
| S1      | C4         | 2 cm   |          | -                | -             |
| S2      | C5         | 2 cm   |          | -                | -             |
| S3      | C3         | 2 cm   |          | -                | -             |
| S4      |            | 26 cm  |          | -                | Cable Ties    |
| S5      | C1         | 2 cm   |          | -                | -             |
| S6      | C6         | 2 cm   |          | -                | -             |
| S7      | C2         | 2 cm   |          | -                | -             |
| S8      | C8, C7     | 260 cm |          | -                | -             |
| S9      | C10, C9    | 260 cm |          | -                | -             |
| S10     | C12, C11   | 260 cm |          | -                | -             |
| S11     | C14, C13   | 260 cm |          | -                | -             |
| S12     | C15, C32   | 260 cm |          | -                | -             |
| S13     | C17, C19   | 12 cm  |          | -                | -             |
| S14     | F18, C20   |        |          | -                | -             |
| S15     | C25, C6    | 2 cm   |          | -                | -             |
| S16     | C23, C21   |        |          | -                | -             |
| S17     | C22, C24   |        |          | -                | -             |
| S18     | C26, C29   | 185 cm |          | -                | -             |
| S19     | C27, C30   | 85 cm  |          | -                | -             |
| S20     | C28, C31   | 45 cm  |          | -                | -             |
| S21     | C16, C32   | 2 cm   |          | -                | -             |

**All the lengths are measured from the BACK OF THE BACKSHELL**

### Pinout Scheme



### Pinout Tables

|                                   |     |    |                                    |     |    |
|-----------------------------------|-----|----|------------------------------------|-----|----|
| C7-C8 (Path 5)<br>Supertronic PVC |     |    | C9-C10 (Path 6)<br>Supertronic PVC |     |    |
| #1                                |     |    | #1                                 |     |    |
|                                   | 1-1 | A+ |                                    | 1-1 | A+ |
|                                   | 2-2 | A- |                                    | 2-2 | A- |
|                                   | 3-3 | B+ |                                    | 3-3 | B+ |
|                                   | 4-4 | B- |                                    | 4-4 | B- |

|                                     |     |    |                                     |     |      |
|-------------------------------------|-----|----|-------------------------------------|-----|------|
| C11-C12 (Path 7)<br>Supertronic PVC |     |    | C13-C14 (Path 8)<br>Supertronic PVC |     |      |
| #1                                  |     |    | #1                                  |     |      |
|                                     | 1-1 | A+ |                                     | 2-2 | GND  |
|                                     | 2-2 | A- |                                     | 1-1 | LED+ |
|                                     | 3-3 | B+ |                                     |     |      |
|                                     | 4-4 | B- |                                     |     |      |

|                                                                |     |  |                                                                    |            |  |
|----------------------------------------------------------------|-----|--|--------------------------------------------------------------------|------------|--|
| C17-C19 (Path 10)<br>4Core 26AWG White PTFE Foil Screened      |     |  | F18-C20 (Path 11)<br>4Core 26AWG White PTFE Foil Screened          |            |  |
| #1 __ screening C17: to pins (4) __ screening C19: to pins (4) |     |  | #1 __ screening F18: not connected __ screening C20: not connected |            |  |
|                                                                | 1-1 |  |                                                                    | tinned 1-1 |  |
|                                                                | 4-4 |  |                                                                    | tinned 2-2 |  |
|                                                                | 2-2 |  |                                                                    | tinned 3-3 |  |
|                                                                | 3-3 |  |                                                                    | tinned 4-4 |  |

|                                      |     |  |                                      |     |    |
|--------------------------------------|-----|--|--------------------------------------|-----|----|
| C26-C29 (Path 15)<br>Supertronic PVC |     |  | C27-C30 (Path 16)<br>Supertronic PVC |     |    |
| #1                                   |     |  | #1                                   |     |    |
|                                      | 1-1 |  |                                      | 1-1 | A+ |
|                                      | 2-2 |  |                                      | 2-2 | A- |
|                                      | 3-3 |  |                                      | 3-3 | B+ |
|                                      | 4-4 |  |                                      | 4-4 | B- |

|                                      |     |    |
|--------------------------------------|-----|----|
| C28-C31 (Path 17)<br>Supertronic PVC |     |    |
| #1                                   |     |    |
|                                      | 1-1 | A+ |
|                                      | 2-2 | A- |
|                                      | 3-3 | B+ |
|                                      | 4-4 | B- |

Based on the attached customer specific cable assembly documentation we confirm the design correctness, freeze of the design, and approve the production of the cable assembly

Responsible technician:

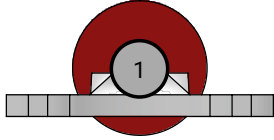
Reviewed and approved by:

\_\_\_\_\_ . date\_\_\_\_\_

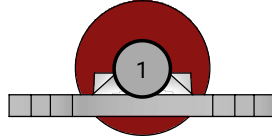
### Pin View (Front View)



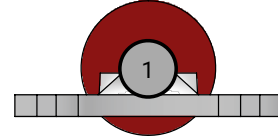
**C1 (N\_PSU):**



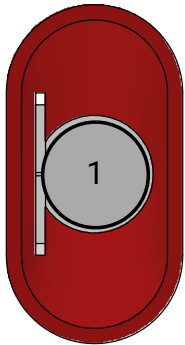
**C3 (PE\_PSU):**



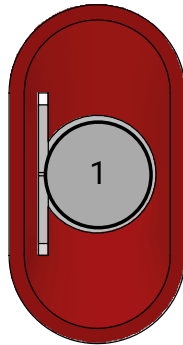
**C2 (F\_PSU):**



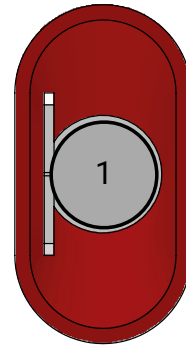
**C4:**



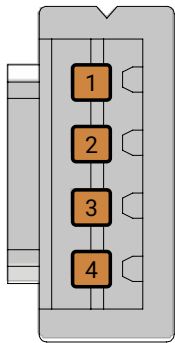
**C5:**



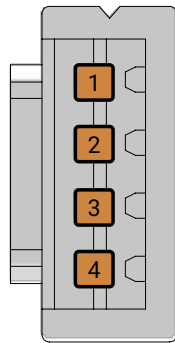
**C6:**



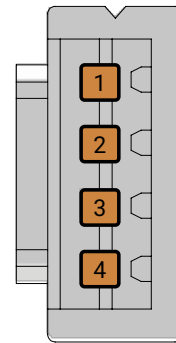
**C7 (Z\_HEAD): (Connected 4)**



**C8 (Z\_CORE): (Connected 4)**

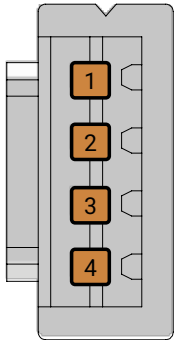


**C9 (L\_HEAD): (Connected 4)**

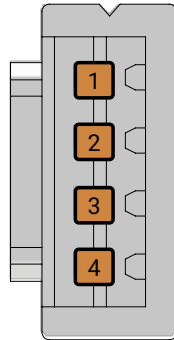




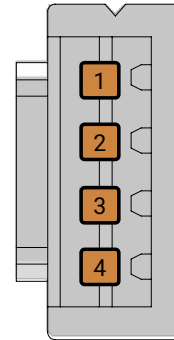
**C10 (L\_CORE): (Connected 4)**



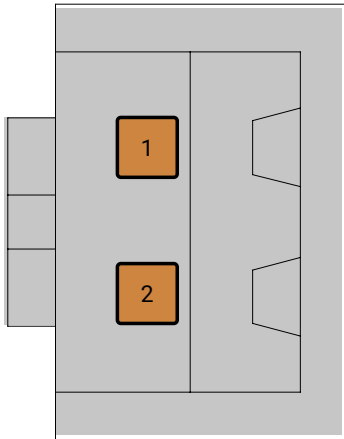
**C11 (R\_HEAD): (Connected 4)**



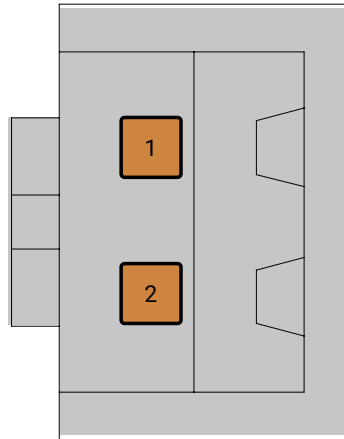
**C12 (R\_CORE): (Connected 4)**



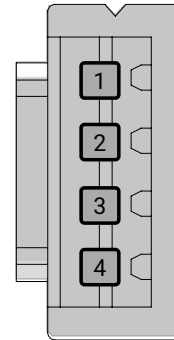
**C14 (LED\_CORE): (Connected 2)**



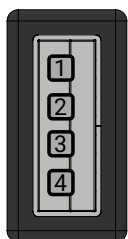
**C13 (LED\_HEAD): (Connected 2)**



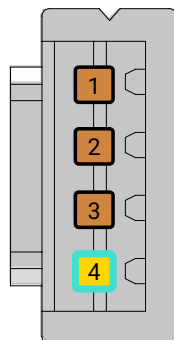
**C15 (USB\_HEAD):**



**C16 (USB\_PLUG\_CAM):**

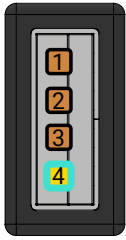


**C17 (USB\_BOT): (Connected 4)**

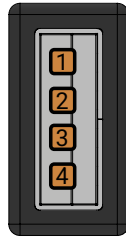


**F18:**

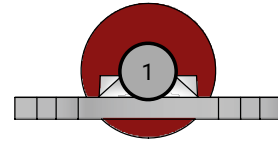
**C19 (USB\_PLUG\_BOT):**  
(Connected 4)



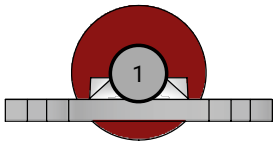
**C20 (USB\_PLUG\_FEEDER):**  
(Connected 4)



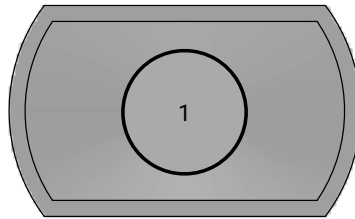
**C21 (24V\_PSU):**



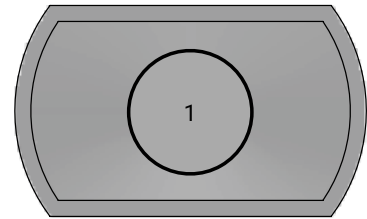
**C22 (GND\_PSU):**



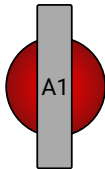
**C23 (24\_IN):**



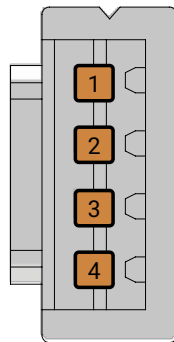
**C24 (GND\_IN):**



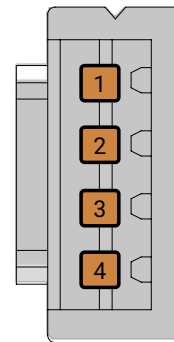
**C25 (FRAME\_PE):**



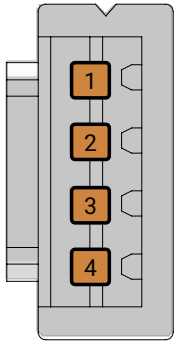
**C26 (X\_PLUG):** (Connected 4)



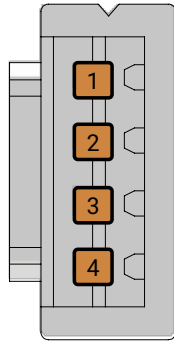
**C27 (Y1\_PLUG):** (Connected 4)



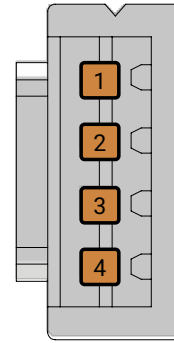
**C28 (Y2\_PLUG):**  
(Connected 4)



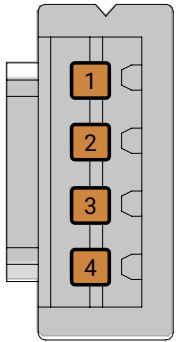
**C29 (X\_CORE):** (Connected 4)



**C30 (Y1\_CORE):**  
(Connected 4)



**C31 (Y2\_CORE):**  
(Connected 4)



**C32:**

